CLASS X MATHEMATICS • CHAPTER 14

Probability

"Exhaustive point-by-point presentation covering theoretical probability, equally likely outcomes, complementary events, cards, dice, and geometric probability directly from NCERT jemh114.pdf."

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Empirical vs Theoretical (Classical) Probability

Recall from Class IX: Empirical Probability

P(E) = (Number of trials in which event happened) / (Total number of trials). Dependent on past experimental repetition, which is unfeasible for earthquakes or satellite launches!

Unbiased & Equally Likely Outcomes

A coin or die is **fair / unbiased** when there is no physical reason to favour any single outcome. **Equally likely outcomes** have the exact same individual chance of occurring (e.g. 1 through 6 on a die).

THEORETICAL (CLASSICAL) PROBABILITY FORMULA

P(E) = n(E) / n(S)

Number of outcomes favourable to E / Number of all possible outcomes

Historical Origin

Defined formally by **Pierre-Simon Laplace in 1795** in his monumental work *Théorie analytique des probabilités* (1812).

⚡ Quick Box • Theoretical Probability Foundations

Q1 What is the key limitation of empirical (experimental) probability that makes theoretical probability necessary?

CONCEPTUAL SHIFT

Q2 If a bag contains 5 red balls and 1 blue ball, are drawing a red ball and drawing a blue ball equally likely outcomes? Explain.

EQUALLY LIKELY

Q3 Who is credited with giving the classical definition of theoretical probability in 1795?

HISTORY CHECK

Pages 205–206 (Textbook)

Elementary Events & Complementary Probability

Elementary Event

An event having only **one single outcome** of the experiment is called an elementary event (e.g., getting a head when tossing a coin once).

SUM OF ELEMENTARY PROBABILITIES

Sum of all elementary event probabilities = 1

Complementary Events (Not E)

The complement of event E, denoted as **P(not E) or P(Ē)**, represents the event failing to occur:

$$P(E) + P(\bar{E}) = 1 \Rightarrow P(\bar{E}) = 1 - P(E)$$

Probability Bounds

For any event E: $0 \le P(E) \le 1$. An impossible event has probability 0; a sure (certain) event has probability 1.

⚡ Quick Box • Events & Probability Bounds

Q1 What is an elementary event? Give an example from a single die throw. — DEFINITION RECALL

Q2 If P(E) = 0.73, what is the probability of "not E"? Can an event have a probability of 1.05? — COMPLEMENT CHECK

Q3 What are the numerical probability values assigned to an impossible event and a sure event respectively? — EXTREMES

Standard Deck of 52 Playing Cards Anatomy

4 Suits of 13 Cards Each

- ✓ **Spades (♠) & Clubs (♣):** 26 Black cards.
- ✓ **Hearts (♥) & Diamonds (♦):** 26 Red cards.
- ✓ **Ranks in each suit:** Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, King.

Face Cards & Honors

- ✓ **Face Cards:** Kings, Queens, and Jacks (3 per suit × 4 suits = **12 face cards** total).
- ✓ **Red Face Cards:** 6 cards (Hearts & Diamonds).
- ✓ **Black Face Cards:** 6 cards (Spades & Clubs).
- ✓ **Aces:** 4 cards total (one per suit).

⚡ Quick Box • Playing Cards Knowledge

Q1 How many total face cards are there in a standard well-shuffled pack of 52 playing cards? — CARD RECALL

Q2 What is the probability of drawing a red king from a standard deck? — PROBABILITY SETUP

Q3 Which suits are black in color, and how many cards belong to each black suit? — SUIT BREAKDOWN

Pages 212–213 (Textbook)

Throwing Two Dice Simultaneously (36 Outcomes)

Sample Space Size

When a blue die and a grey die are thrown together (or one die twice):

Total Possible Outcomes = 6 × 6 = 36

Ordered pairs (x, y) where x is blue die, y is grey die. Note that (1, 4) and (4, 1) are distinct outcomes!

Sum of Two Numbers Analysis

- ✓ **Minimum sum:** 1 + 1 = 2 (Probability = 1/36).
- ✓ **Maximum sum:** 6 + 6 = 12 (Probability = 1/36).
- ✓ **Most frequent sum:** Sum = 7 has 6 outcomes — (1,6),(2,5),(3,4),(4,3), (5,2),(6,1) (Probability = 6/36 = 1/6).
- ✓ **Sum = 13:** Impossible event (Probability = 0).

⚡ Quick Box • Two Dice Sample Space

Q1 What is the total number of outcomes when two dice are thrown simultaneously? TOTAL SIZE

Q2 How many ordered pairs yield a sum of exactly 8 when throwing two dice? FAVORABLE COUNT

Q3 Why is getting a sum of 13 considered an impossible event in a two-dice experiment? EXTREMES

Pages 210–212 (Textbook)

Geometric Probability (Length & Area Ratios)

1D Geometric Probability (Time / Length)

When outcomes lie continuously along a line segment (e.g. musical chair stopping within 2 minutes):

P(E) = (Favorable Length) / (Total Length)

2D Geometric Probability (Area Ratios)

When an object (like a crashed helicopter or dropped die) lands randomly within a 2D region (rectangle, lake):

P(E) = (Favorable Area) / (Total Area)

⚡ Quick Box • Geometric Probability

Q1 Why does the standard discrete formula P(E) = n(E)/n(S) fail when outcomes are continuous numbers or points in a region? — CONCEPT LIMIT

Q2 In a 2D geometric probability problem, what ratio replaces favorable outcomes over total outcomes? — AREA RATIO

Q3 If a music track stops randomly within 2 minutes, what is the probability it stops in the first 30 seconds? — APPLICATION

EXERCISE HIGHLIGHTS • PART 1

Pages 214–215 (Textbook)

Exercise Highlights: Spinner & Card Problems

EXERCISE 14.1 • QUESTION 12 (SPINNER)

Game of Chance (Numbers 1 to 8)

Total outcomes = {1, 2, 3, 4, 5, 6, 7, 8} (n = 8).

- ✓ (i) P(points at 8) = **1/8**.
- ✓ (ii) P(odd number) = {1,3,5,7} → 4/8 = **1/2**.
- ✓ (iii) P(> 2) = {3,4,5,6,7,8} → 6/8 = **3/4**.
- ✓ (iv) P(< 9) = {1..8} → 8/8 = **1 (Sure event)**.

EXERCISE 14.1 • QUESTION 14 (PLAYING CARDS)

One Card from 52 Cards

Total outcomes n = 52.

- ✓ (i) P(king of red colour) = 2/52 = **1/26**.
- ✓ (ii) P(face card) = 12/52 = **3/13**.
- ✓ (iii) P(red face card) = 6/52 = **3/26**.
- ✓ (iv) P(jack of hearts) = **1/52**.
- ✓ (v) P(spade) = 13/52 = **1/4**.

EXERCISE HIGHLIGHTS • PART 2

Pages 215–216 (Textbook)

Exercise Highlights: Defective Pens & Numbered Discs

EXERCISE 14.1 • QUESTION 16

Mixed Pens (12 Defective + 132 Good)

Total pens = 12 + 132 = **144**.

- ✓ We need the probability that a randomly drawn pen is a **good one**.
- ✓ Favorable outcomes (good pens) = **132**.
- ✓ P(good pen) = 132 / 144 = 11 / 12 (dividing numerator and denominator by 12).

P(good pen) = 11 / 12

EXERCISE 14.1 • QUESTION 18

90 Discs Numbered 1 to 90

Total outcomes n = 90.

- ✓ (i) Two-digit numbers (10 to 90) = 81 numbers → P = 81/90 = **9/10**.
- ✓ (ii) Perfect squares ({1,4,9,16,25,36,49,64,81}) = 9 numbers → P = 9/90 = **1/10**.
- ✓ (iii) Divisible by 5 (multiples of 5 up to 90) = 18 numbers → P = 18/90 = **1/5**.

Page 217 (Textbook)

Chapter 14 Master Formula & Concept Checklist

Core Probability Principles

- ✓ **Theoretical Probability:**
  $P(E) = n(E) / n(S)$
- ✓ **Probability Bounds:**
  $0 \leq P(E) \leq 1$
- ✓ **Complementary Rule:**
  $P(E) + P(\text{not } E) = 1$

Key Conceptual Distinctions

- ✓ **Sure Event:** Probability equals 1.
- ✓ **Impossible Event:** Probability equals 0.
- ✓ **Empirical vs Theoretical:** Experimental trials predict what *actually happened*; classical probability predicts what *should happen* based on assumptions.

NCERT CLASS X MATHEMATICS

Thank You!

"Probability is common sense reduced to calculation. Master the science of uncertainty and make informed decisions in a complex world."

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